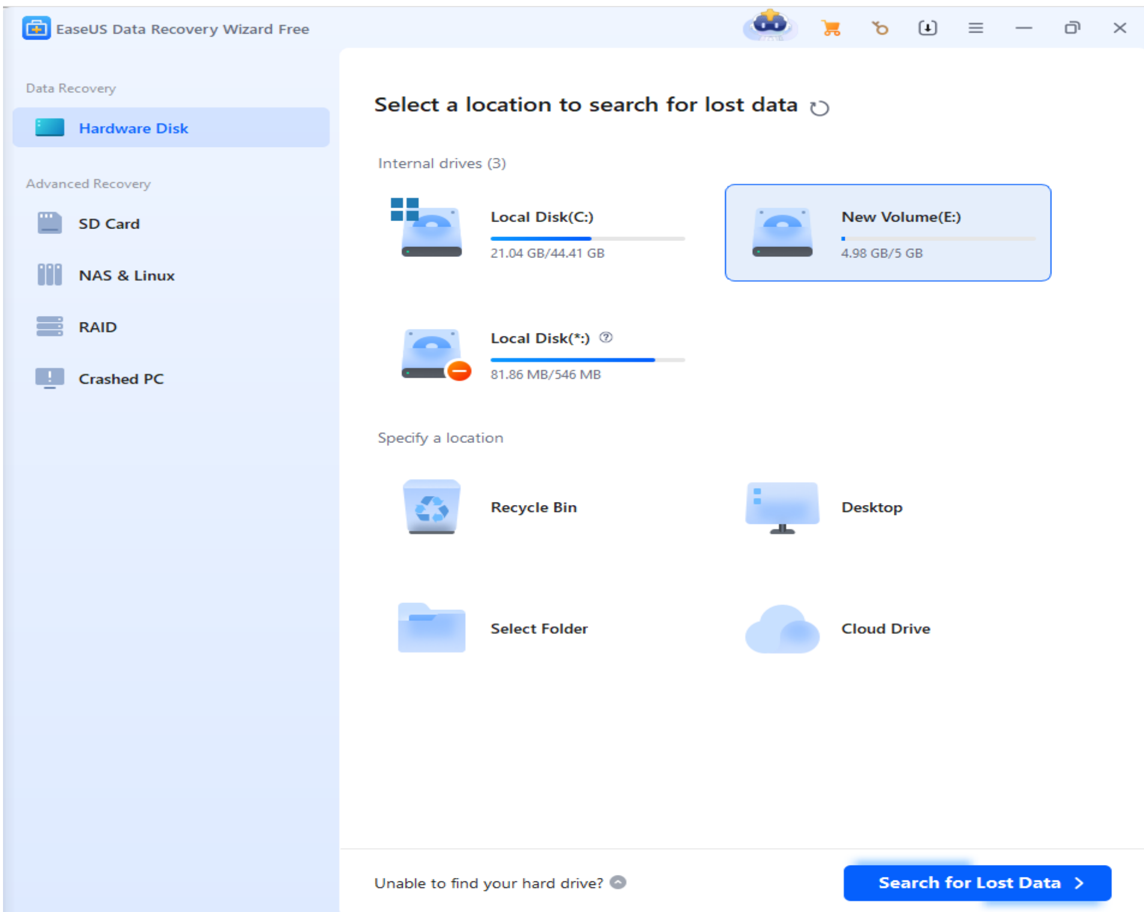
Cyber Forensics
Lab 1 - Computer Forensics Investigation Process

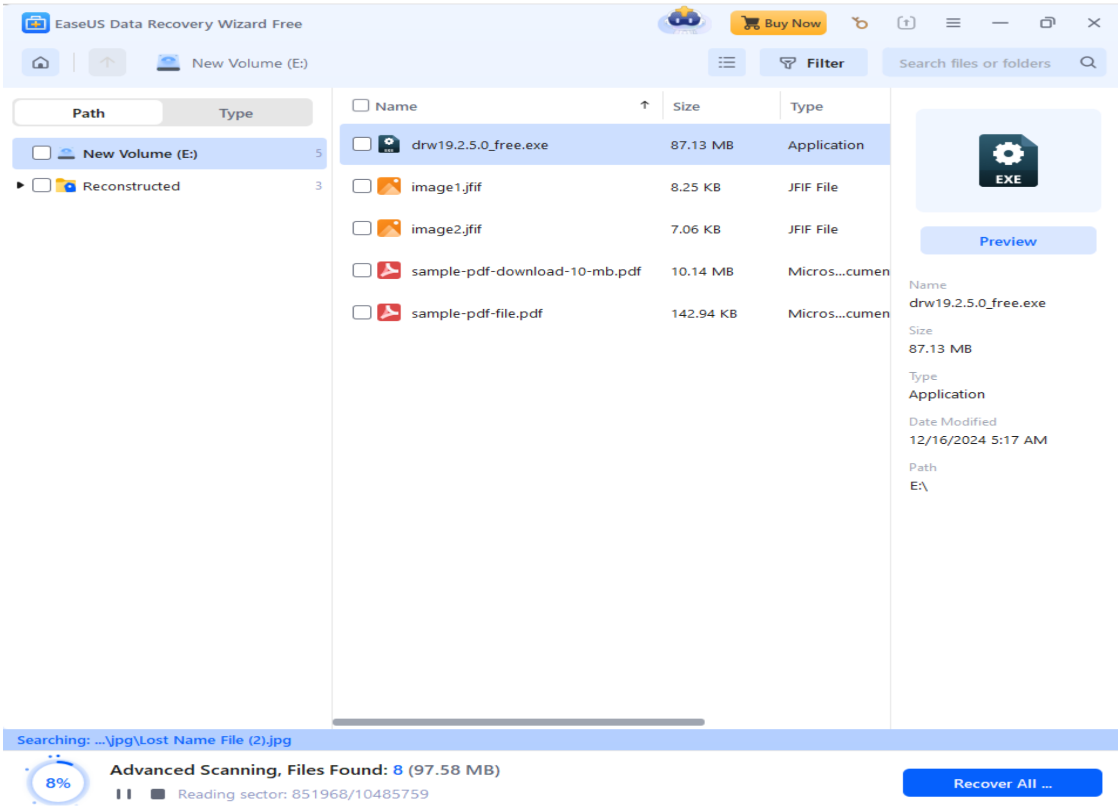
1. Recovering Data Using the EaseUS Data Recovery Wizard

EaseUS Data Recovery Wizard is a popular software designed to recover lost, deleted, or formatted data from various storage devices like hard drives, SSDs, USB drives, and memory cards. It offers an intuitive interface, making it easy for users to recover data without technical expertise. The tool supports multiple file formats and provides options for quick and deep scans to retrieve data efficiently.

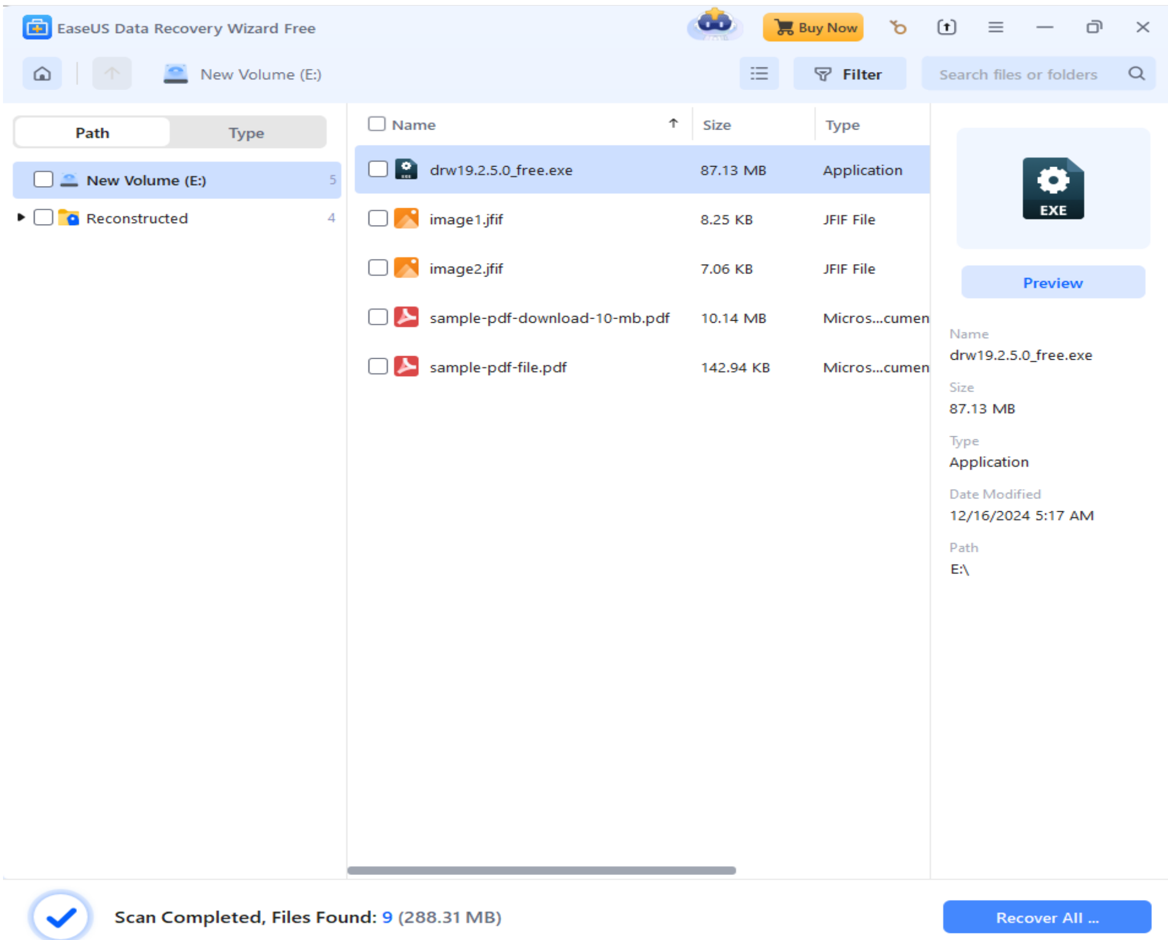
1. Install EaseUS Data recovery Wizard

2. Open EaseUD and select a drive (Drive E) and after that select Scan

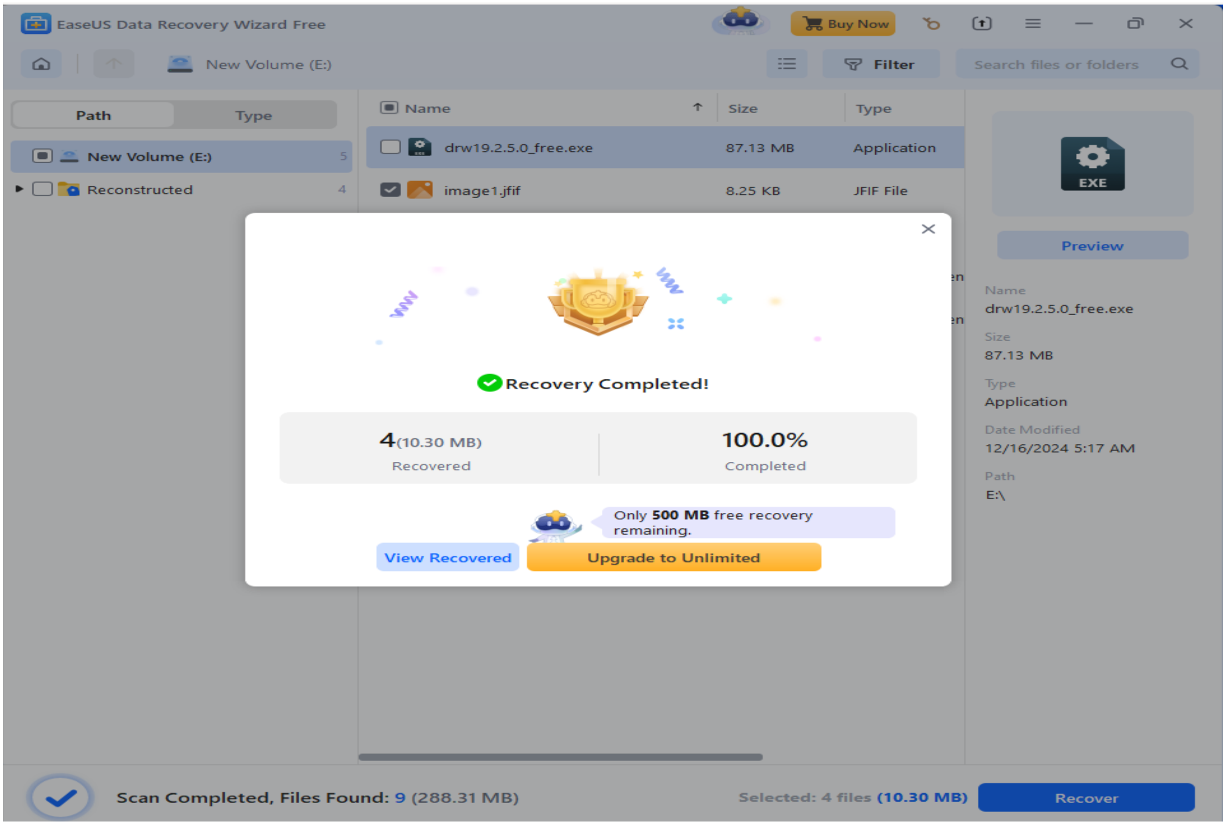




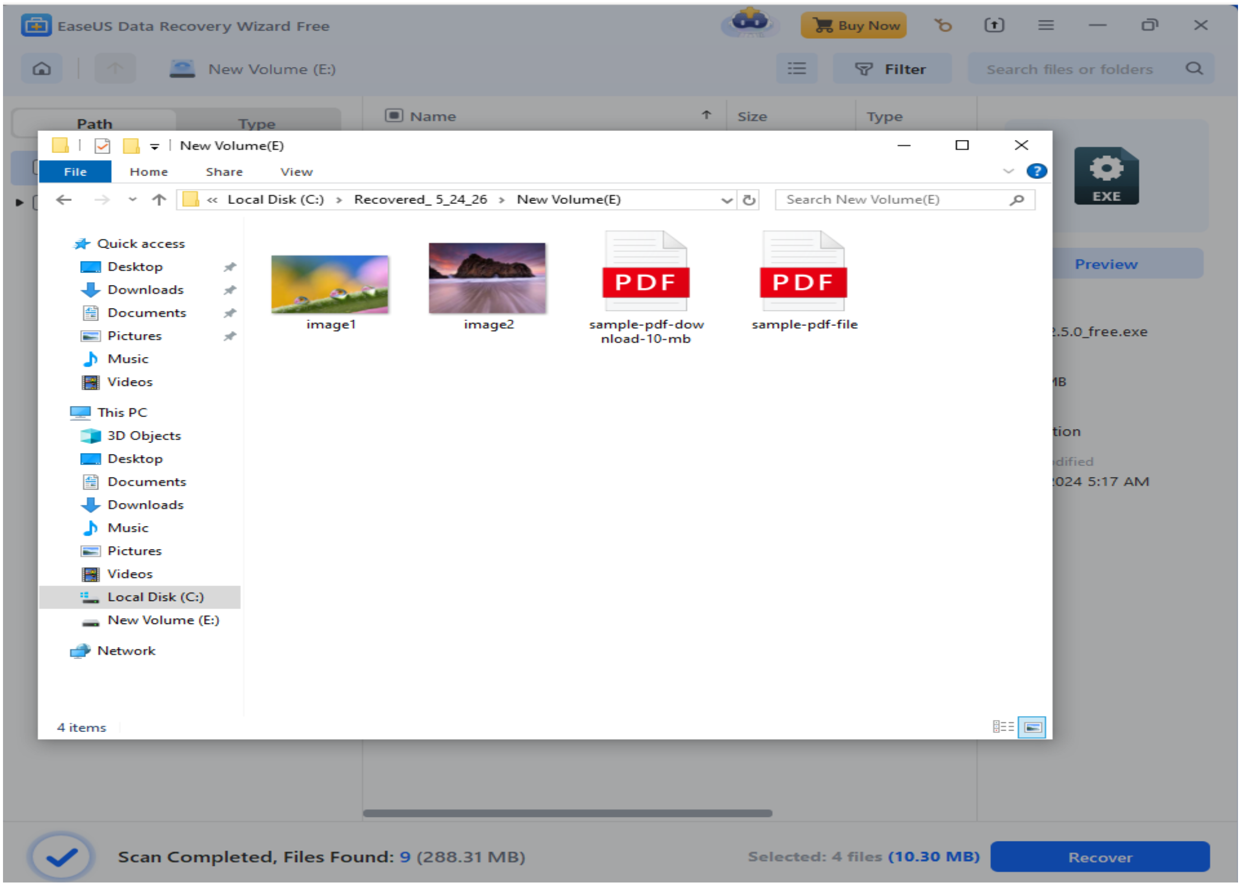
3. The Application will scan the drive and display the contents of the drive along with the data that has been deleted.



4. Then select the lost files which has to be restored and click recover.



5. The files are successfully recovered and stored in a Recovered_5_24_26 folder.

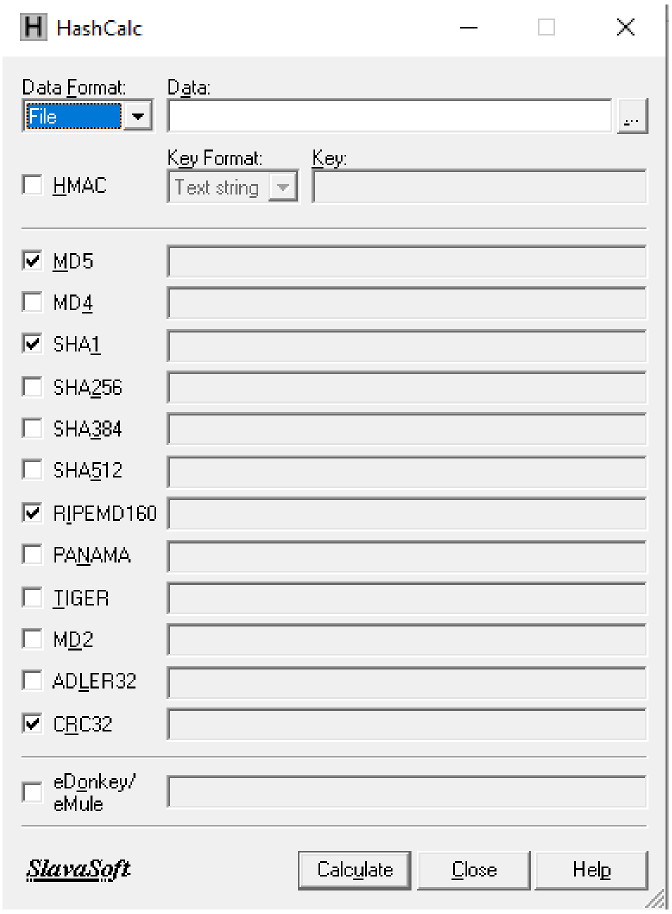


2. Performing Hash, Checksum, or HMAC Calculations Using the HashCalc

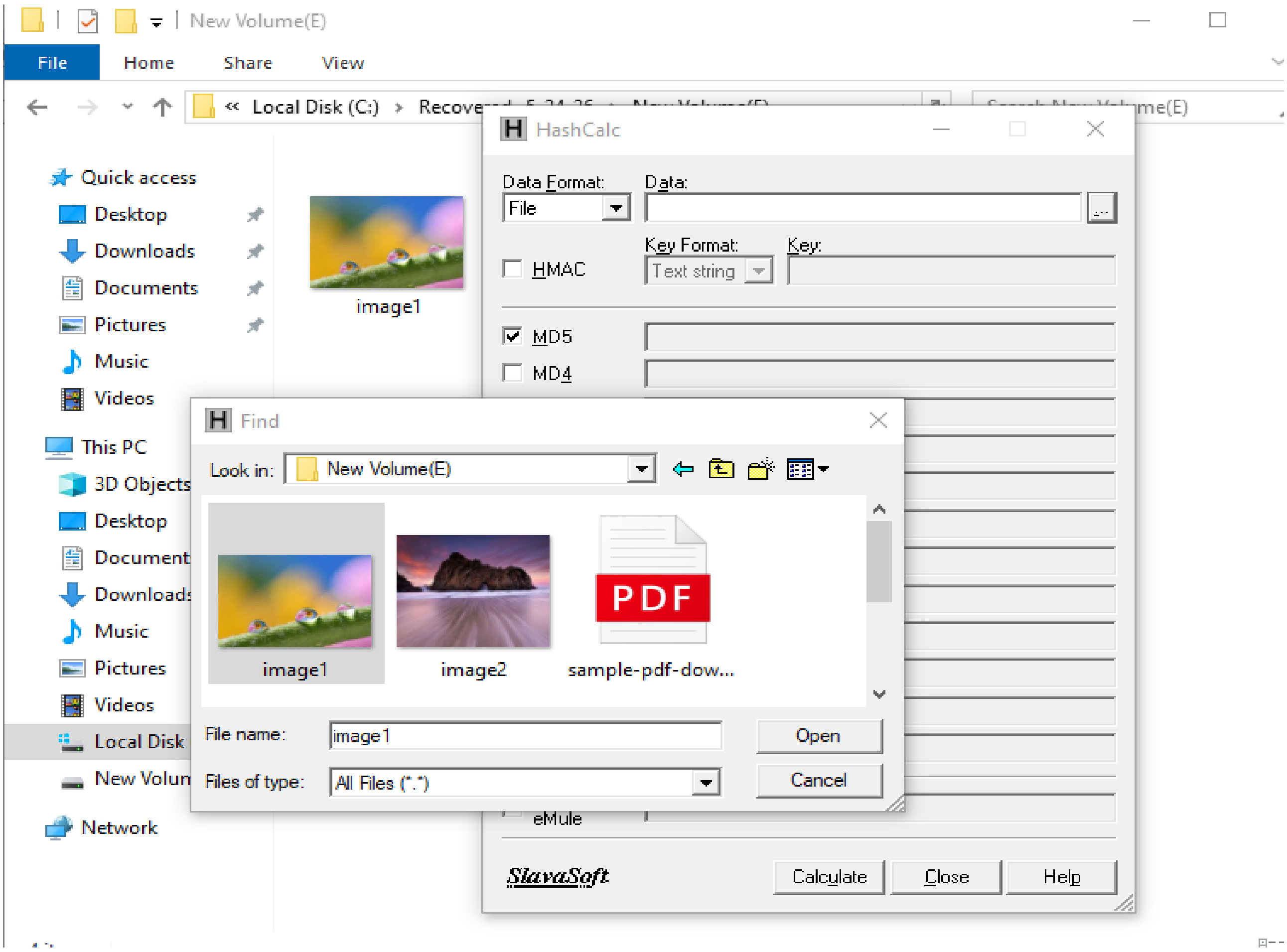
HashCalc is a lightweight, free utility for calculating various hash values, checksums, and HMACs for files, text, and data streams. It supports algorithms like MD5, SHA-1, SHA-256, CRC32, and more, making it ideal for verifying data integrity. The tool features a simple interface, ensuring ease of use for beginners and professionals alike. HashCalc is compatible with Windows systems and is widely used for file verification and cryptographic tasks.

HashCalc allows you to compute gigests, checksums and HMACs for files as well as for text and hex strings.

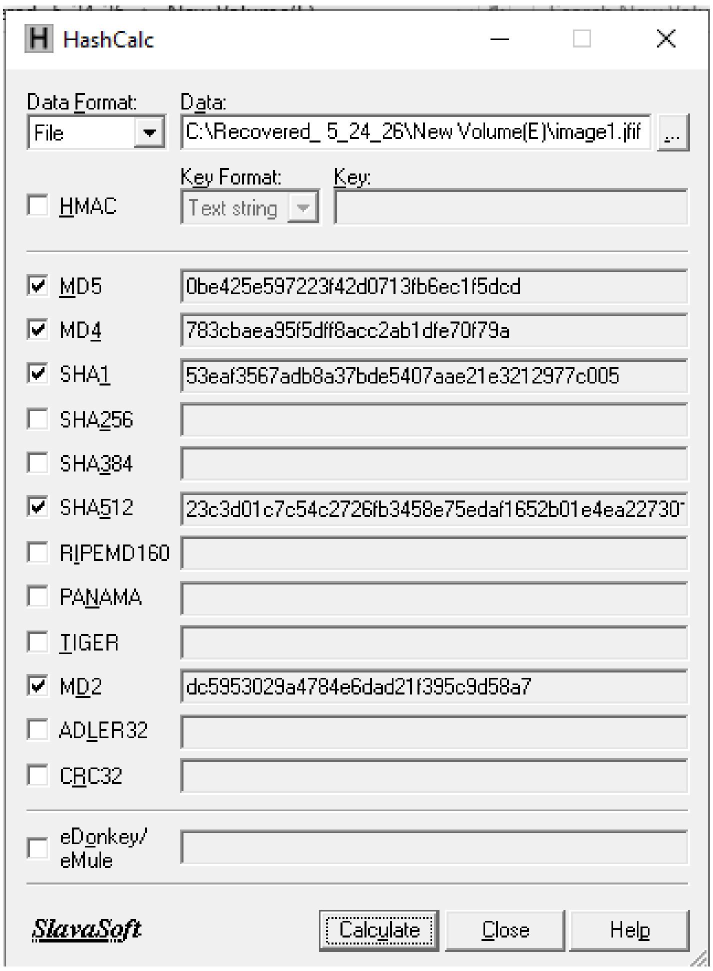
1. Install HashCalc
2. Open HashCalc



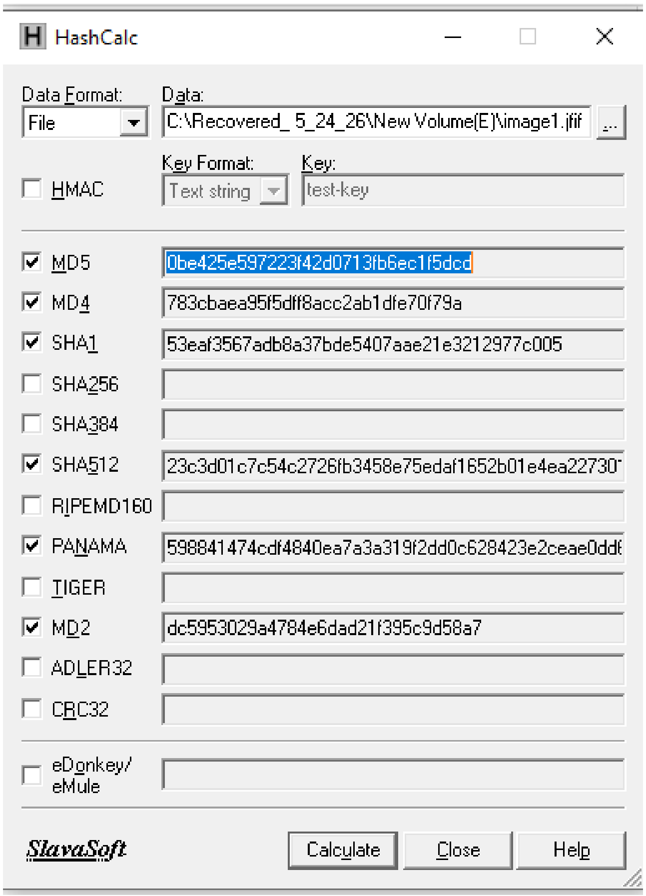
2. In the data format dropdown select File and in data select the file for which the hash has to be calculated.

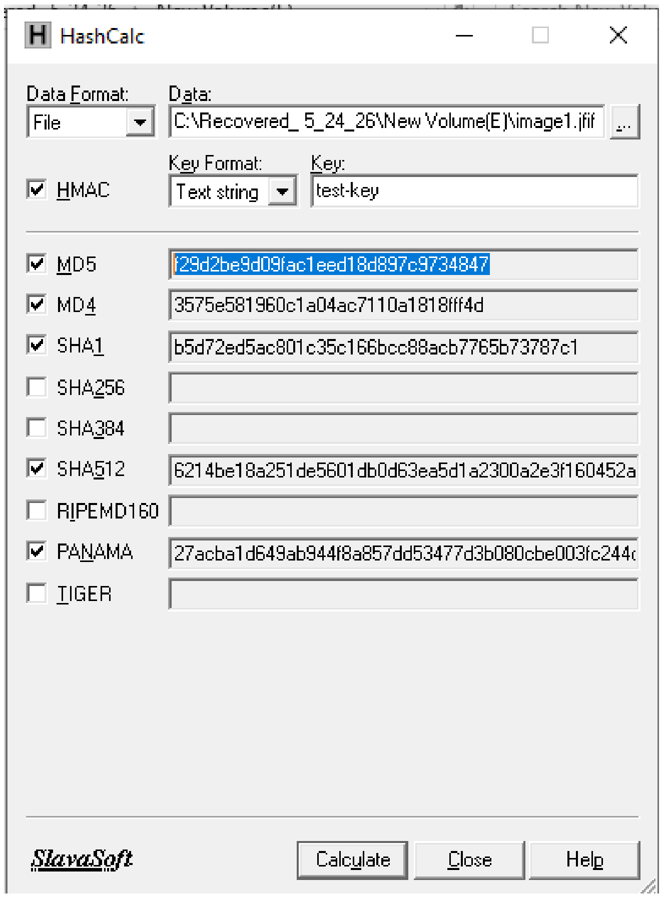


4. The hash values will be displayed for the selected files

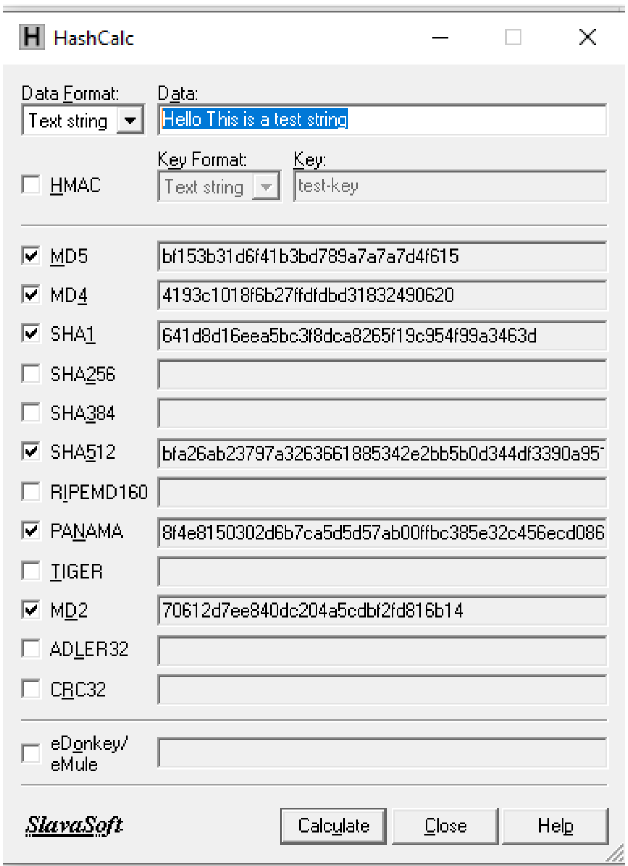


5. Hash can be calculated with key and without key





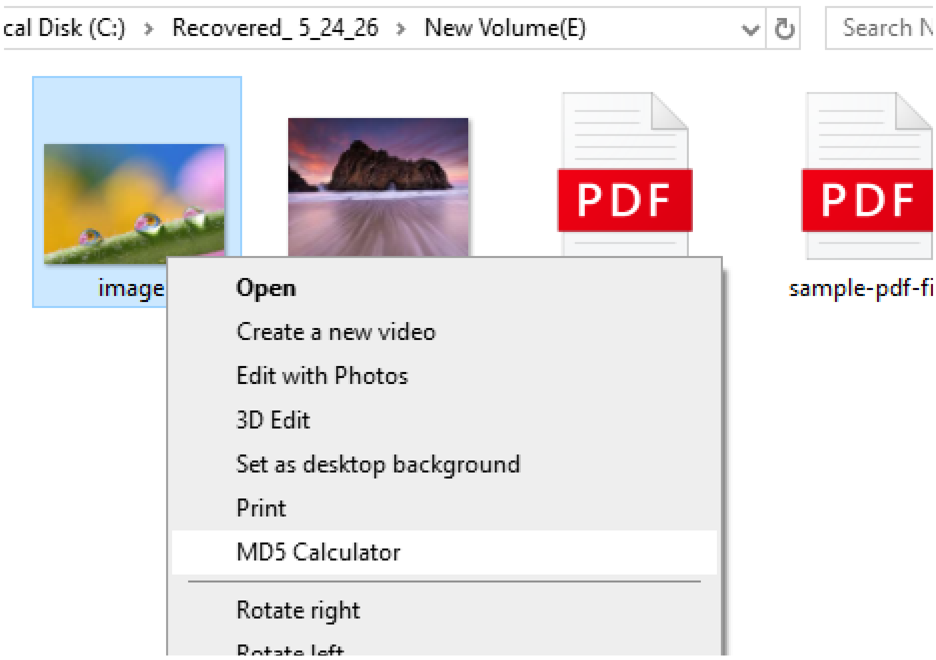
6. If you want to calculate the hash for a text then select text string the data format and provide the text data in data field.



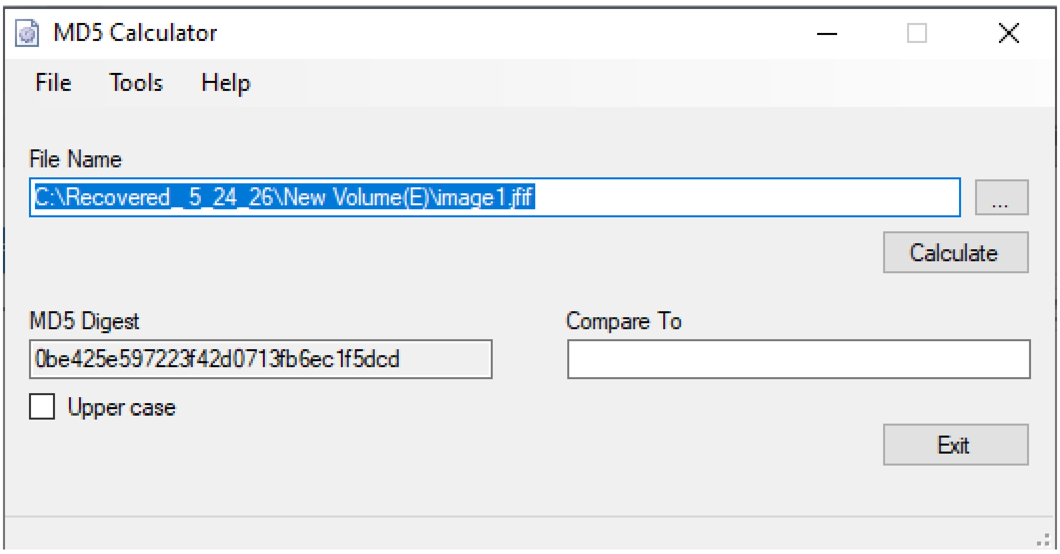
3. Generating MD5 Hashes Using MD5 Calculator

An MD5 Calculator is a tool used to generate the MD5 hash of files or data to verify integrity and authenticity. It plays a key role in digital forensics by detecting file tampering and ensuring evidence remains unaltered. The tool uniquely identifies files, aiding in deduplication and comparison. Despite its utility, MD5 is outdated for security purposes due to vulnerabilities to collisions.

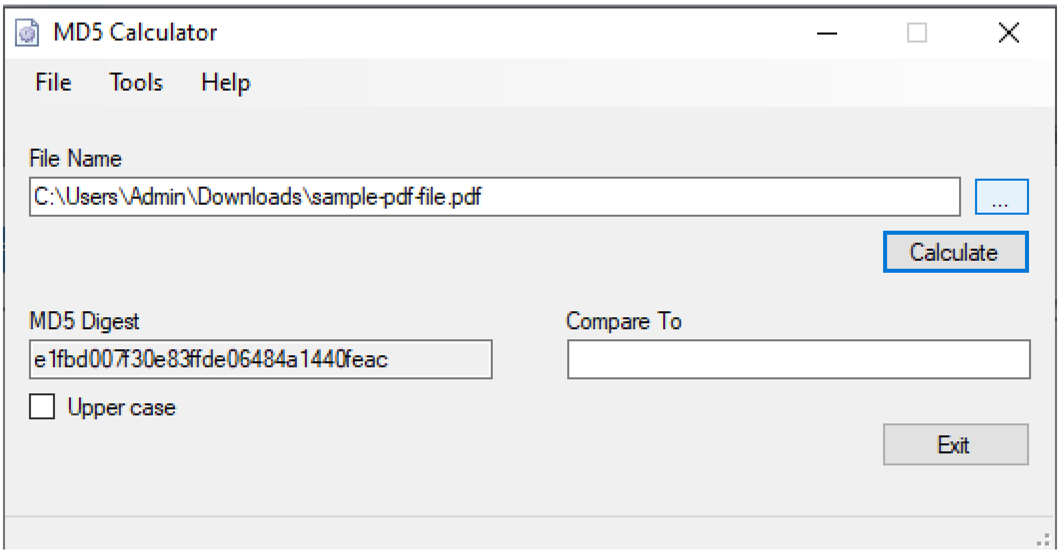
1. Select the file and right click and then select MD5 Calculator.



2. The MD5 calculator window will appear , displaying the MD5 hash value for the selected file.



3. If you want to calculate the hash value of another file click the ellipsis button near the file name field and select the file and click open.



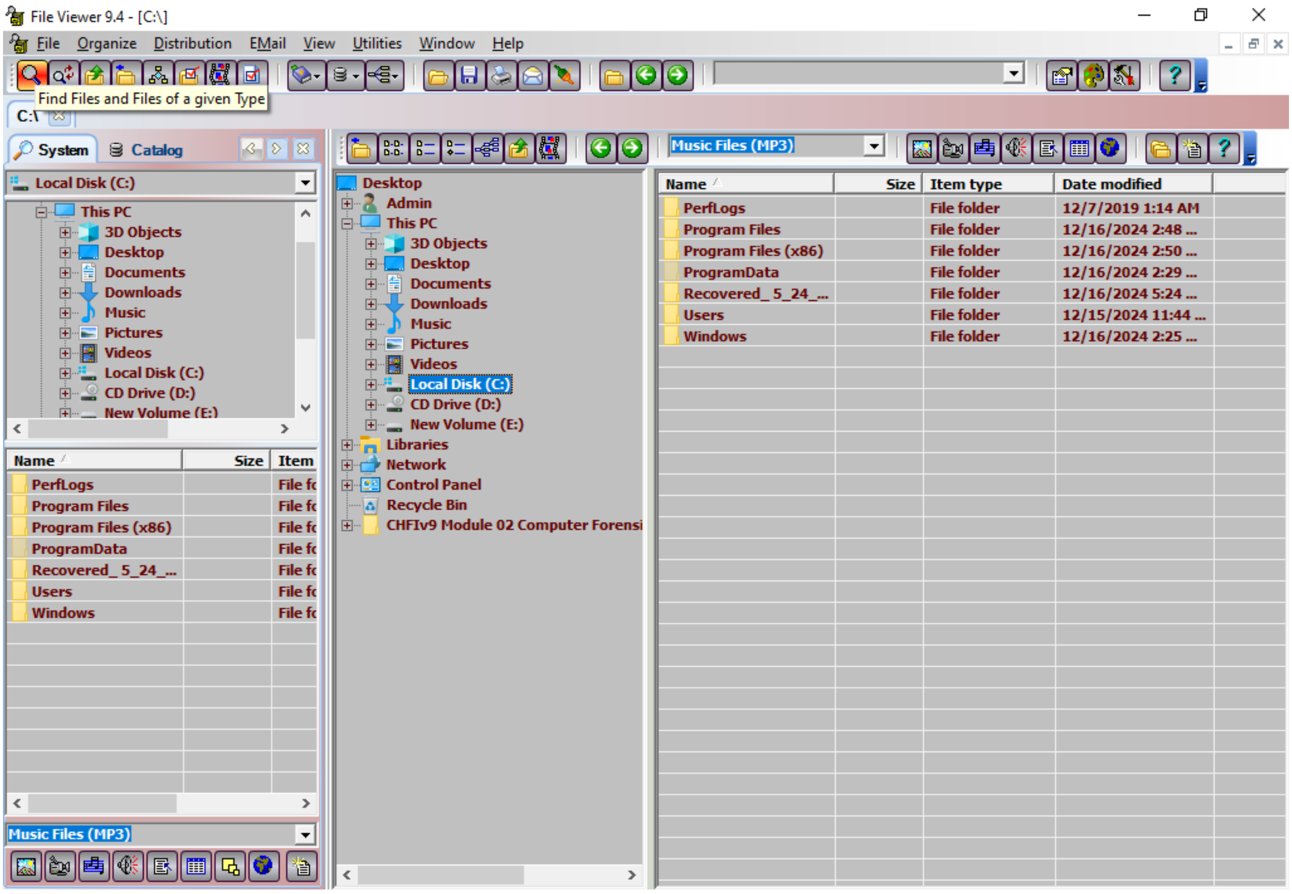
4. Viewing Files of Various Formates Using the File Viewer

File Viewer is a Disk/File Utility that helps you quickly locate, view, print, organize, and exchange files.

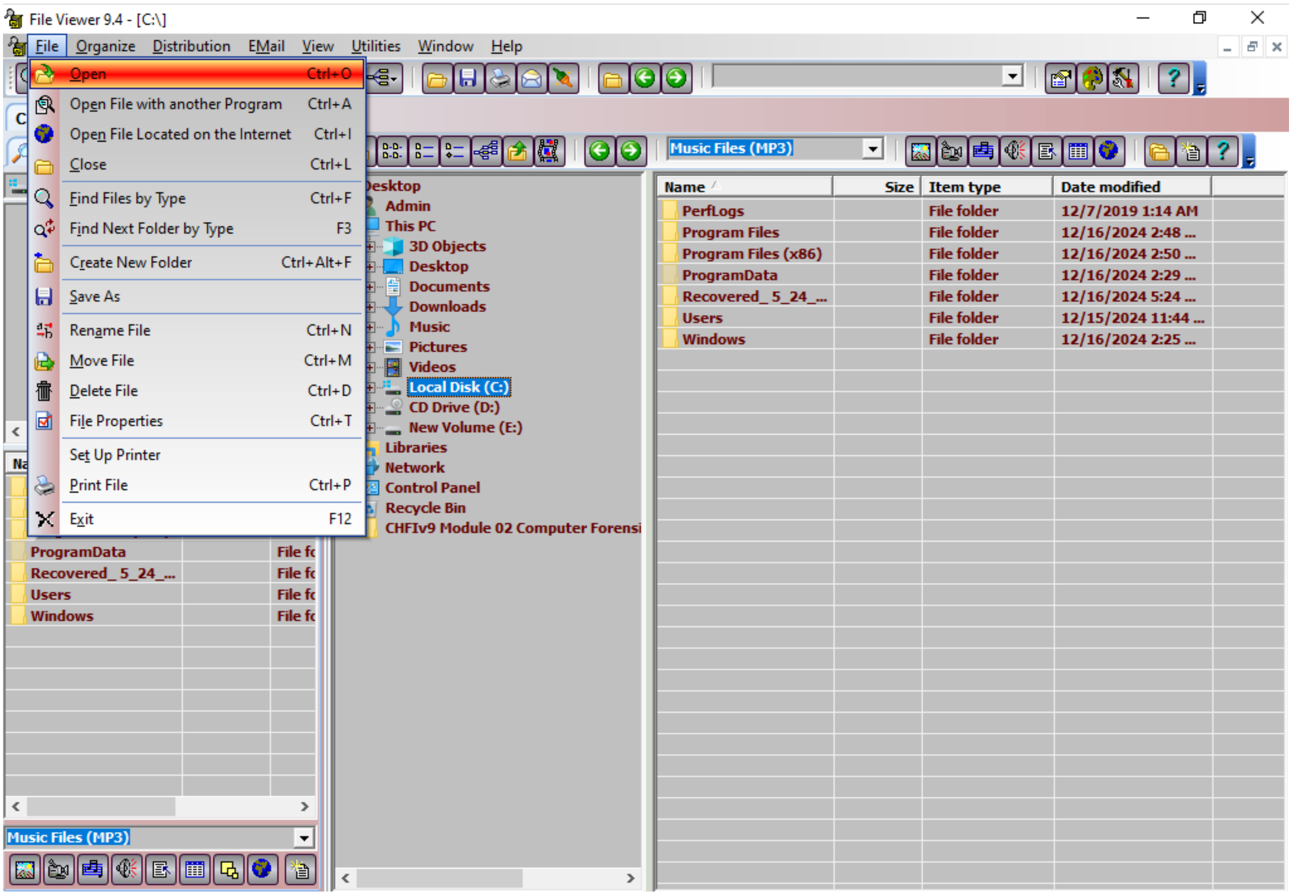
File viewer is used for:

- Viewing files of various formats
- Quickly locating the files needed
- Saving files of various file types

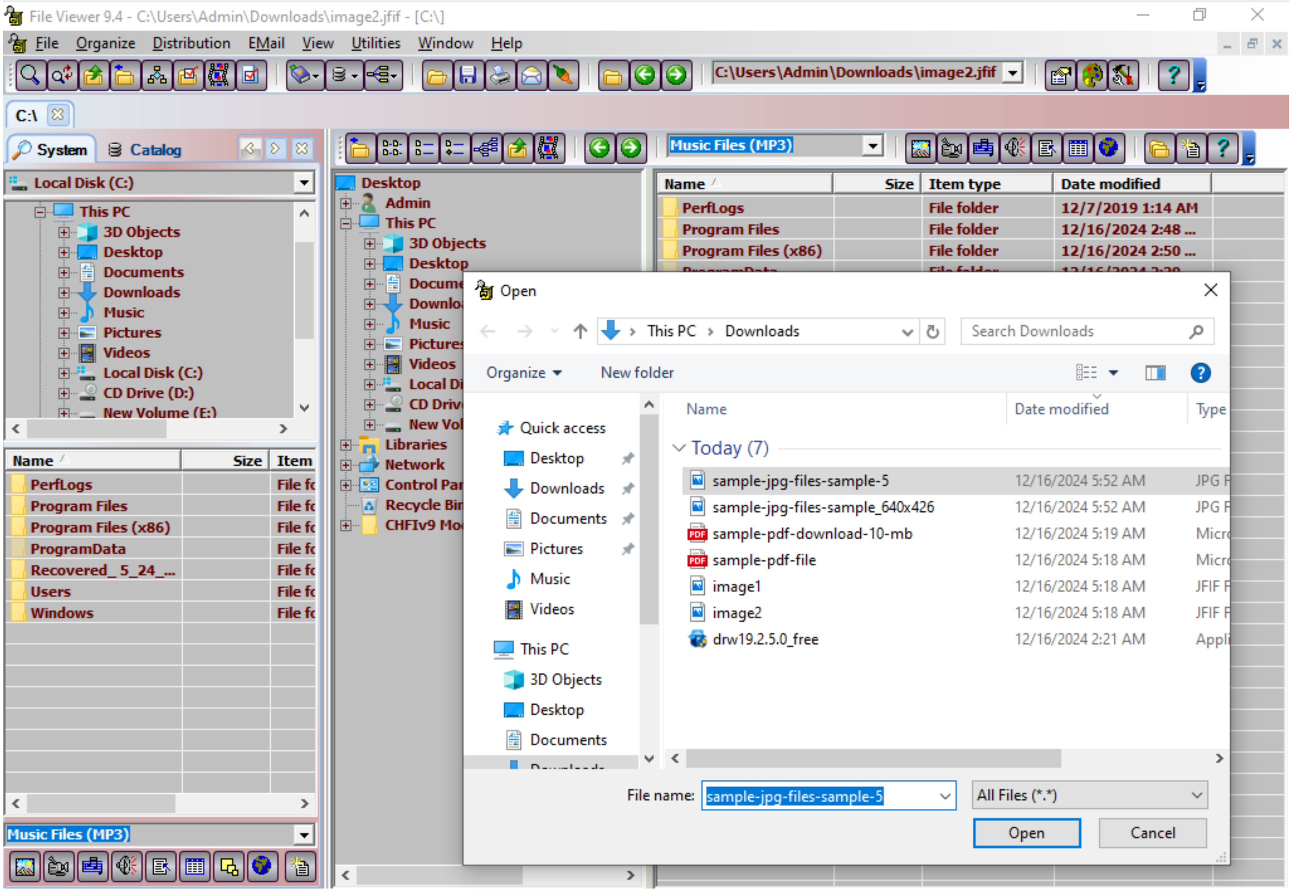
1. install and open file viewer.



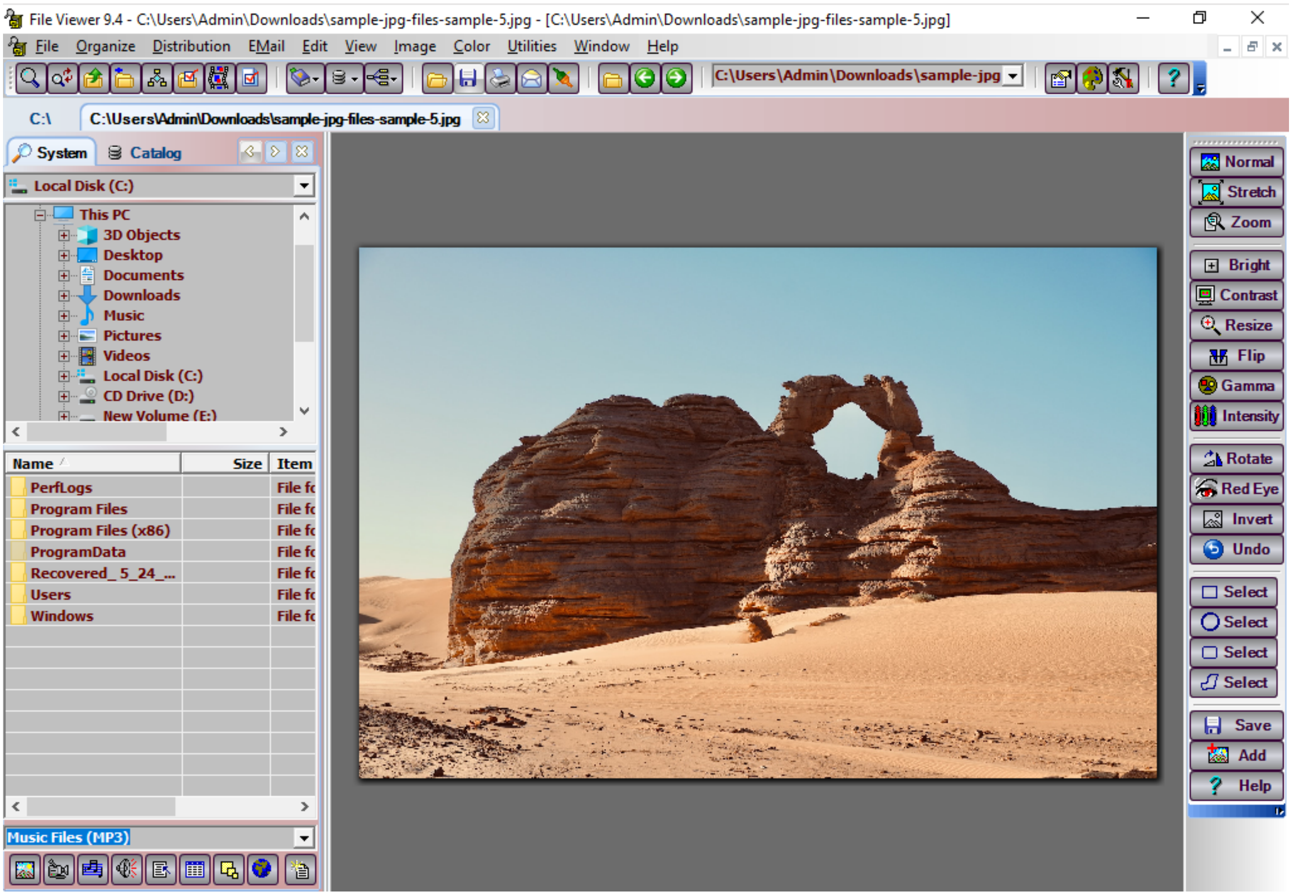
2. Go to File menu and click Open



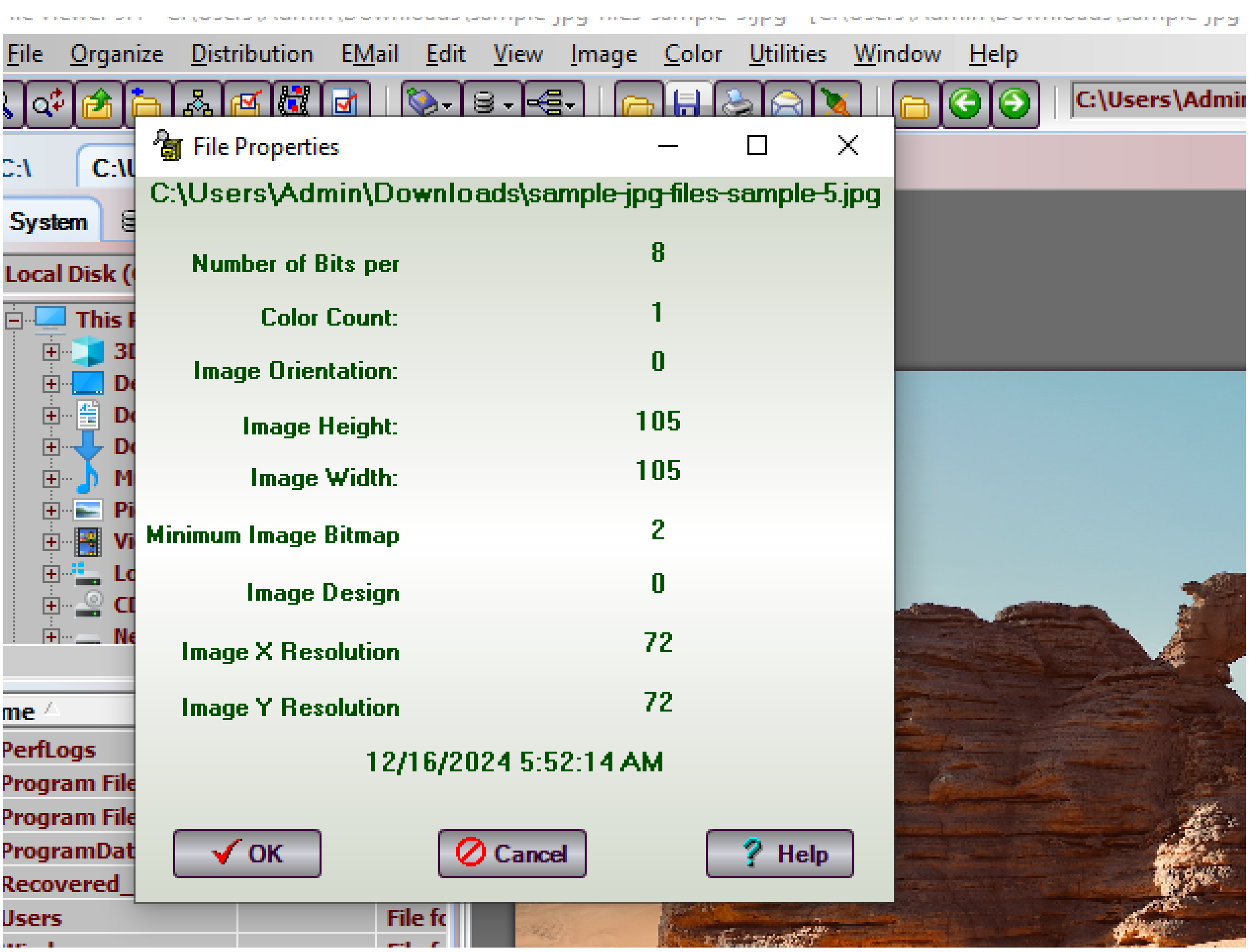
3. Select the file and click open



3. The selected image file opens in the file viewer screen as shown below.



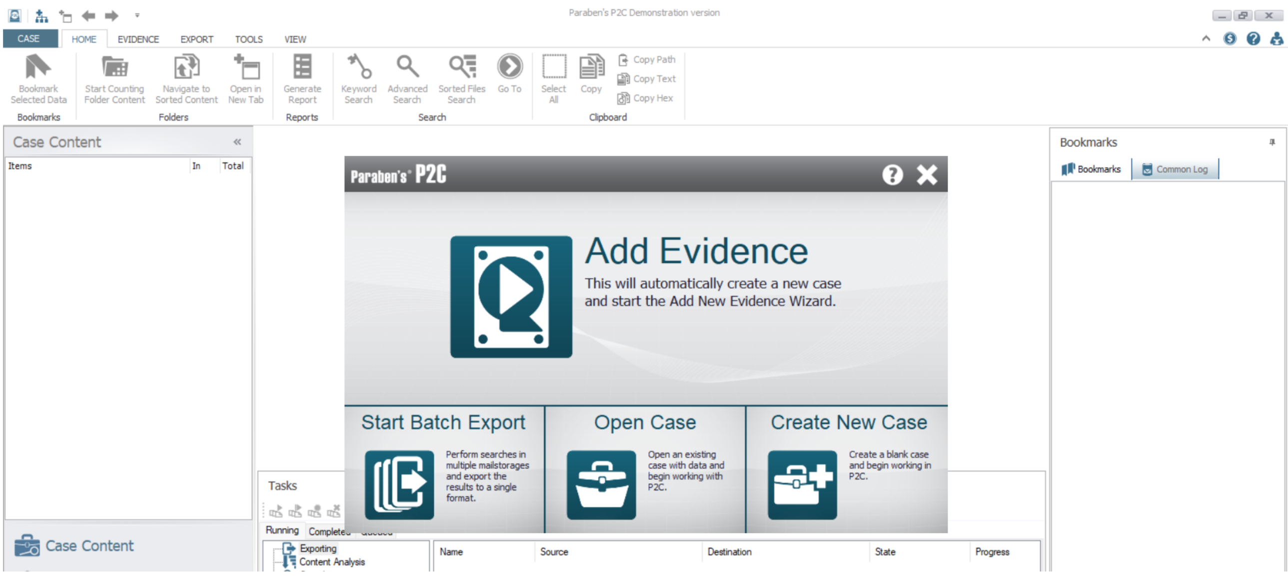
4. Navigate to File -> File Properties to vie the various properties of the selected image



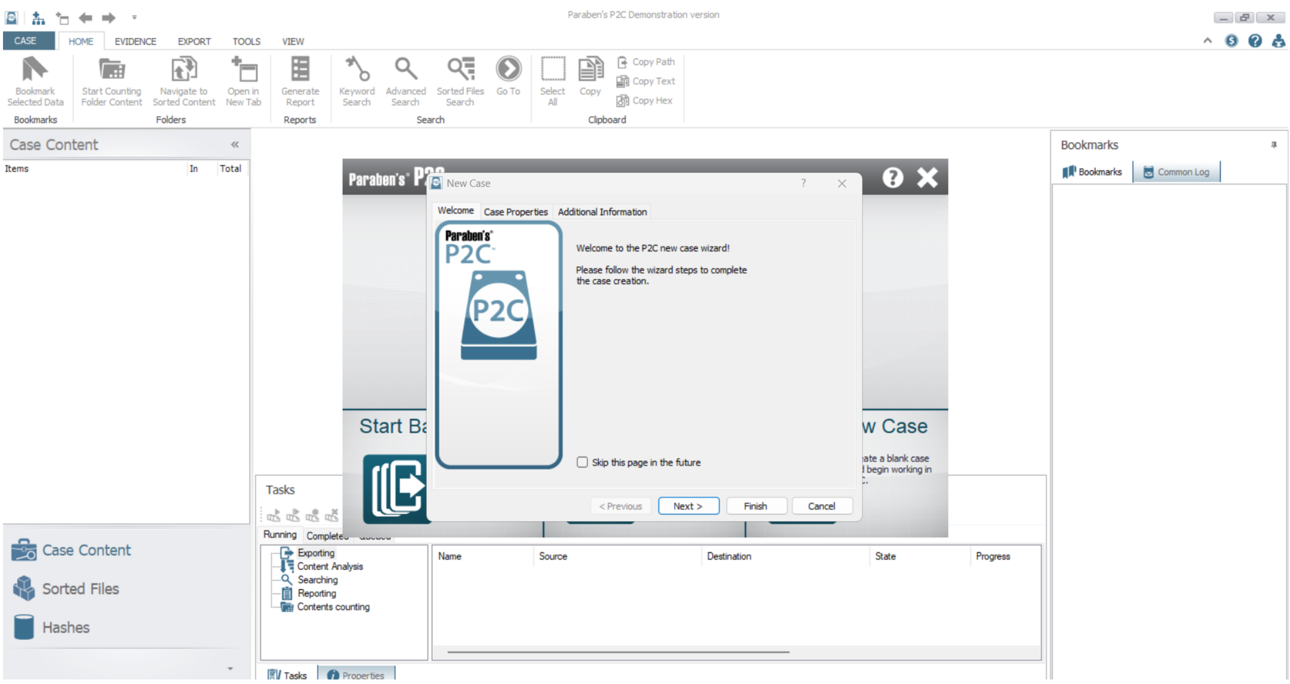
5. Handling Evidence Data Using the P2 Commander

P2 Commander is a comprehensive digital forensic tool designed to handle more data, more efficiently, while keeping a specialized focus on the entire forensic examination process.

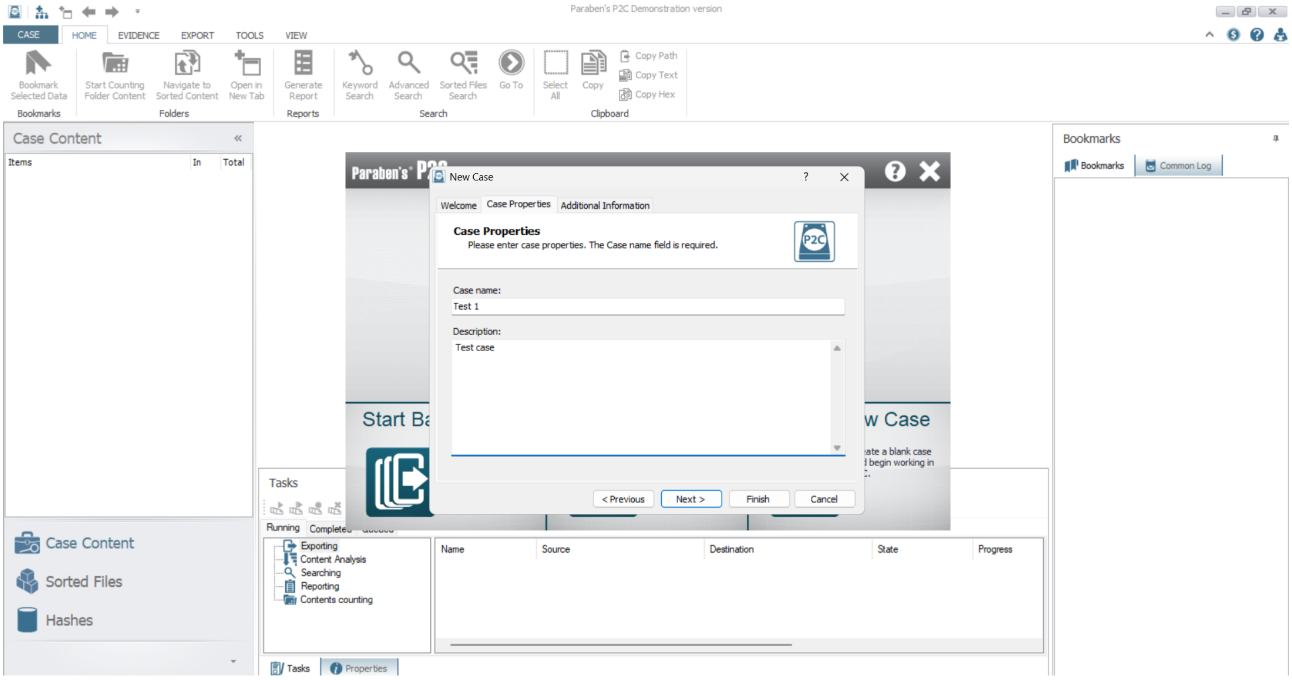
1. Install and open P2 commander
2. Click create new case icon in the Paraben's P2C popup



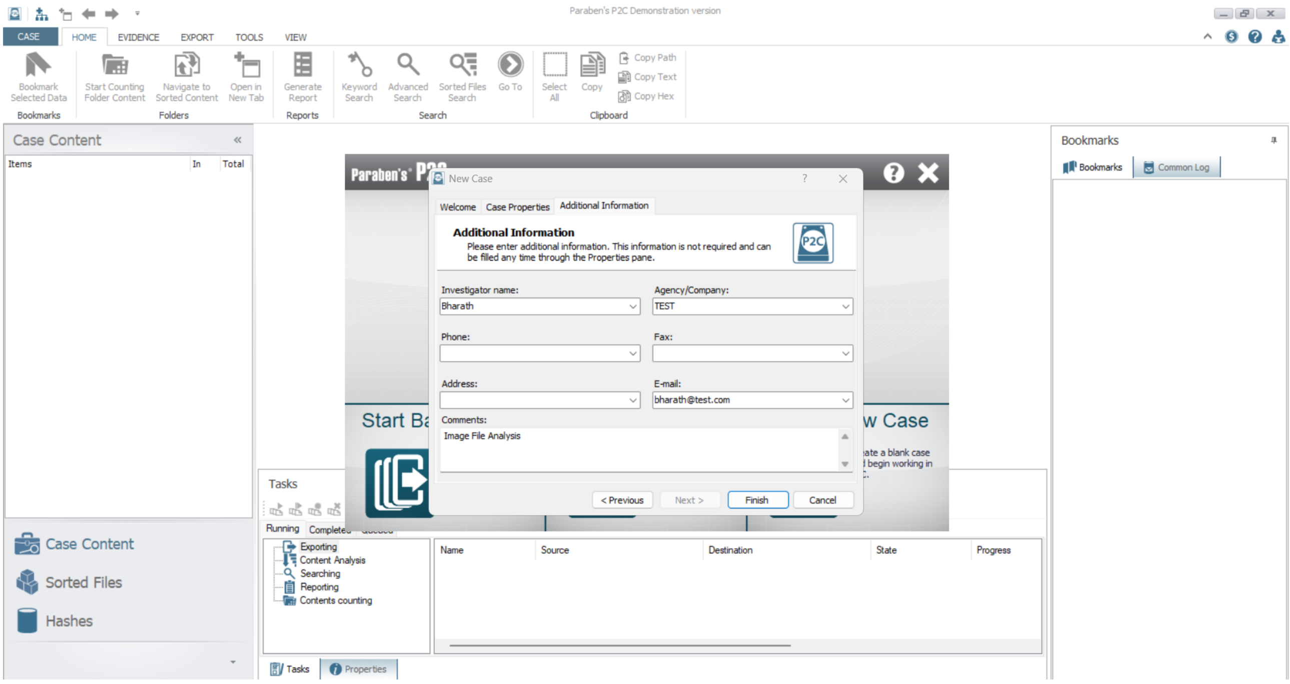
3. A New Case window appears , displaying the welcome section, click Next



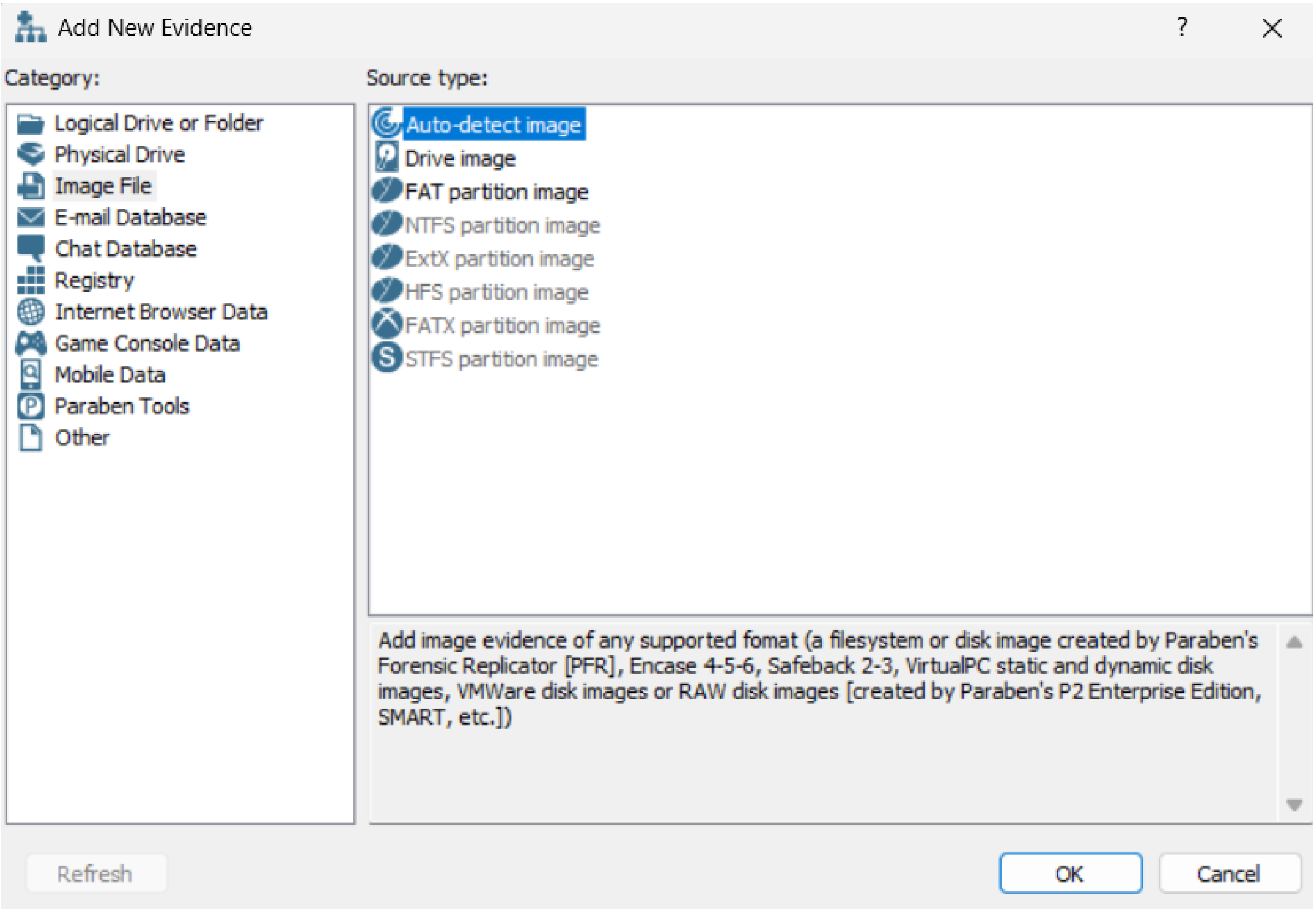
4. In case properties section , provide case name and description and click next



5. In addictional information section fill the additional informations and click finish button



6. In the Add New Evidence window, select Image File under the Category section in the left pane, then select Auto-detect image under the Source type section and click OK.

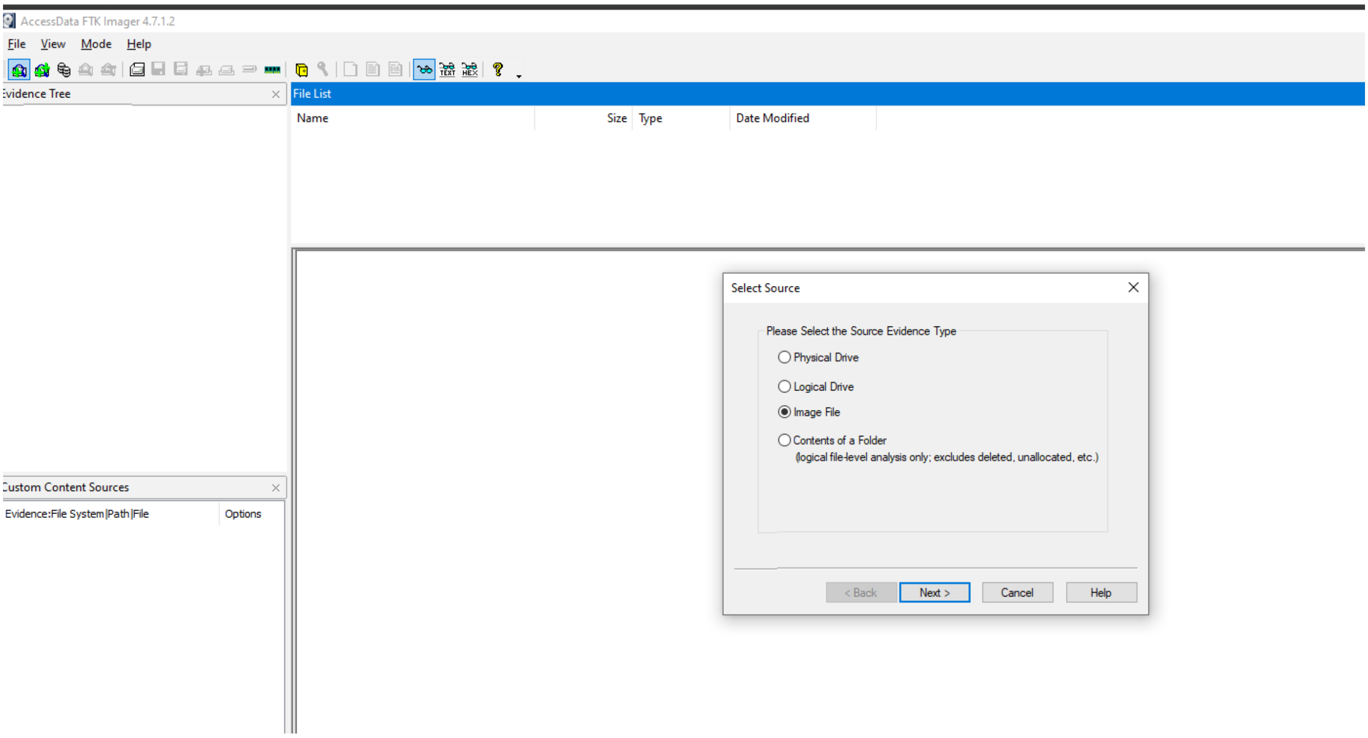


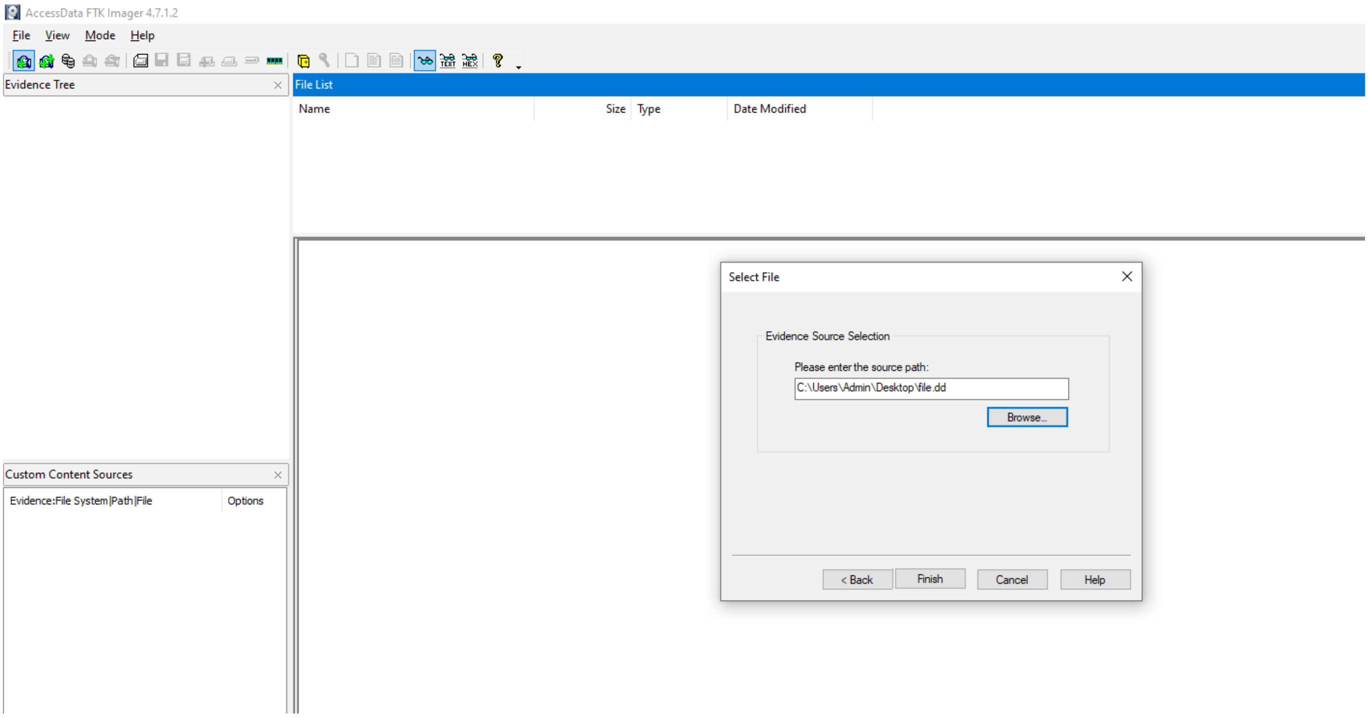
P2C doesn't supported the disk image files. So I proceeded with an alternate tool [FTK Imager]

FTK Imager

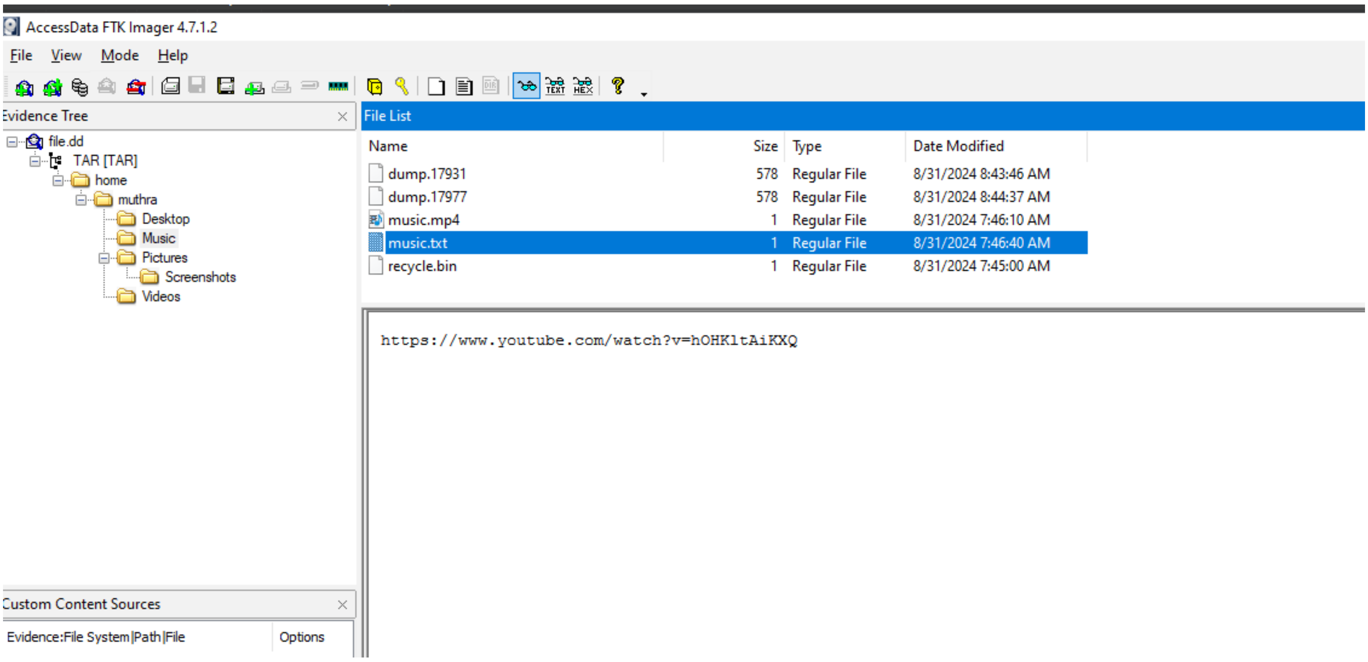
FTK Imager is a forensic imaging tool used to acquire, analyze, and preserve digital evidence. It creates exact copies of data from storage devices without altering the original, ensuring integrity for investigation.

1. Install and open FTK Imager
2. Click on add evidence file icon
3.Select image file option and choose the dsik image file (file.dd)


AccessData FTK Imager 4.7.1.2
File View Mode Help
Evidence Tree
File List
Name
Size Type
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
Select Source
Please Select the Source Evidence Type
Physical Drive
Logical Drive
Image File
Contents of a Folder
(logical file-level analysis only; excludes deleted, unallocated, etc.)
< Back
Next >
Cancel
Help


AccessData FTK Imager 4.7.1.2
File View Mode Help
Evidence Tree
File List
Name
Size Type
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
Select File
Evidence Source Selection
Please enter the source path:
C:\Users\Admin\Desktop\file.dd
Browse...
< Back
Finish
Cancel
Help

4. The following image shows the contents in the disk image.



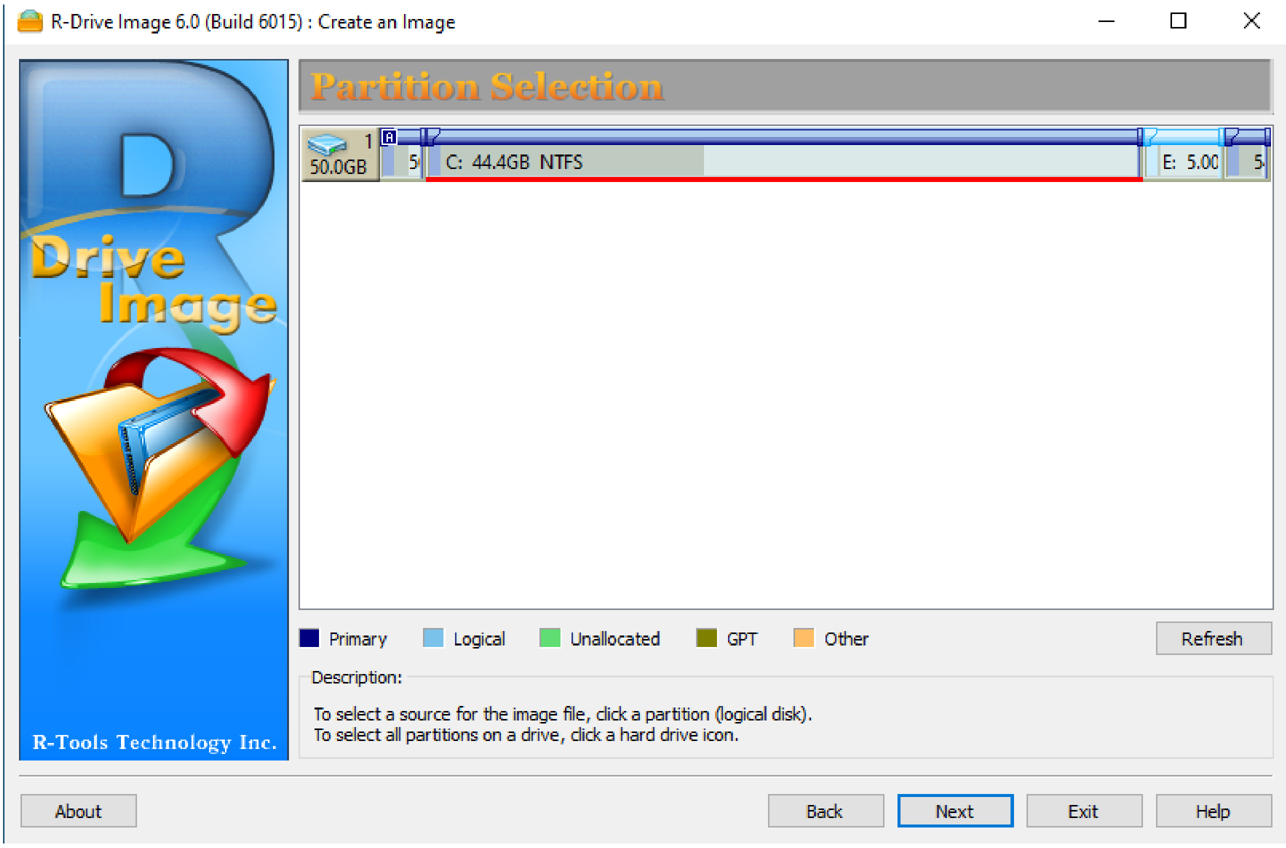
6. Creating a Disk Image file of a hard disk partion using the R-Drive Image

R-Drive Image is a disk imaging software that allows users to create exact copies of hard drives or partitions for backup, recovery, or system migration. It supports a wide range of file systems and provides options for both full and incremental backups. The software ensures data integrity and facilitates quick recovery in case of system failures or data loss.

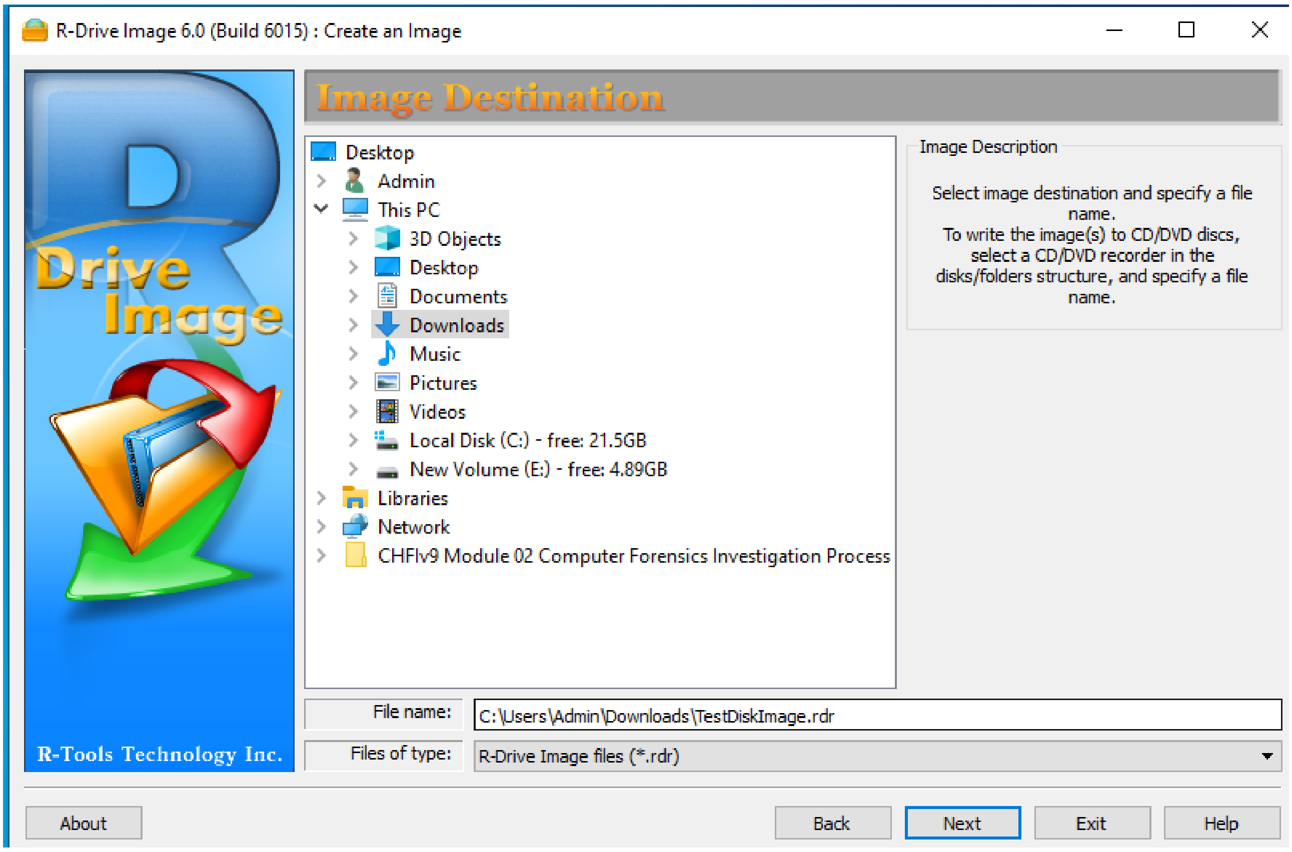
1. Install and open R-Drive image

2. Click Create an image option

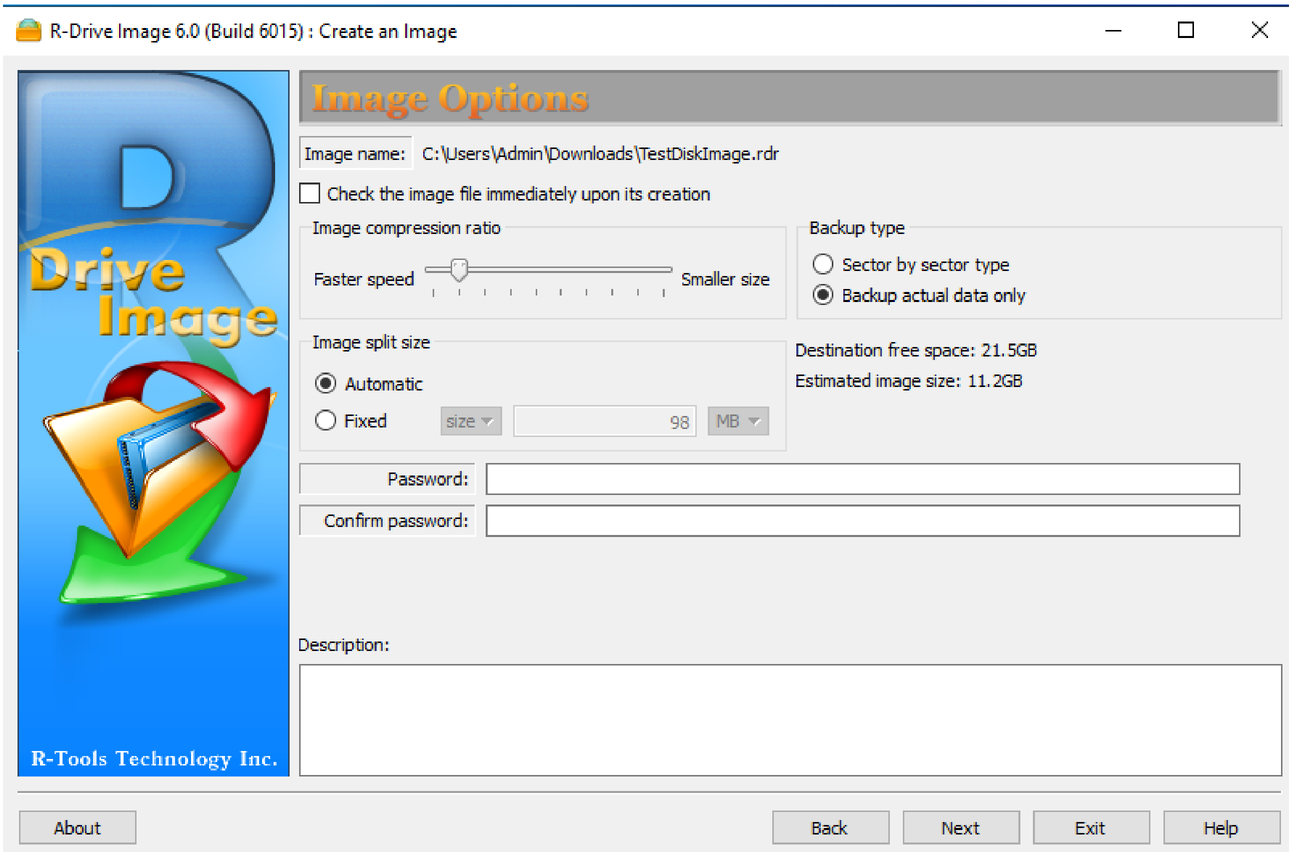
3. In the partition selection window select E drive to create a drive image of the the Drive, click Next.



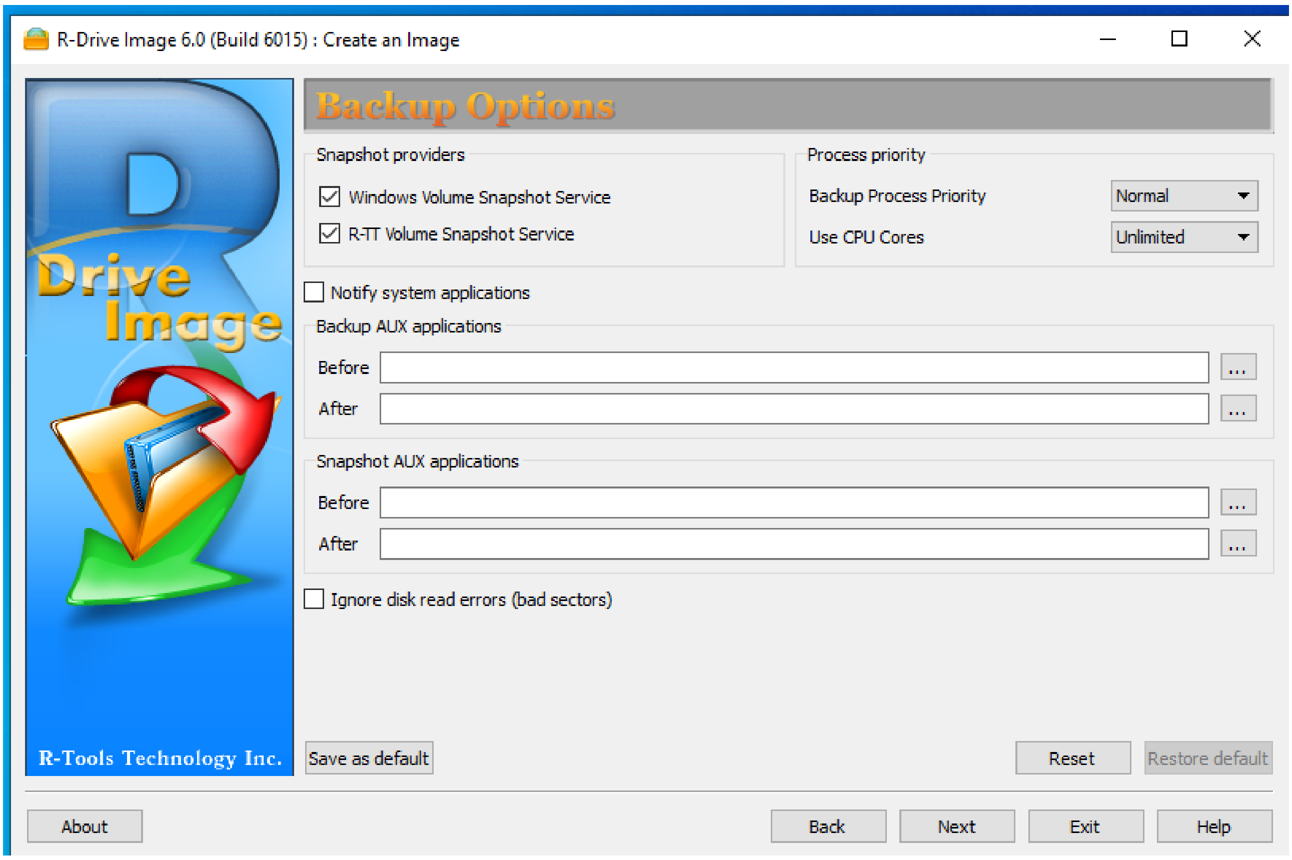
4. Select the folder for saving the file and then select .rdr in files of type field to save the file and click Next.

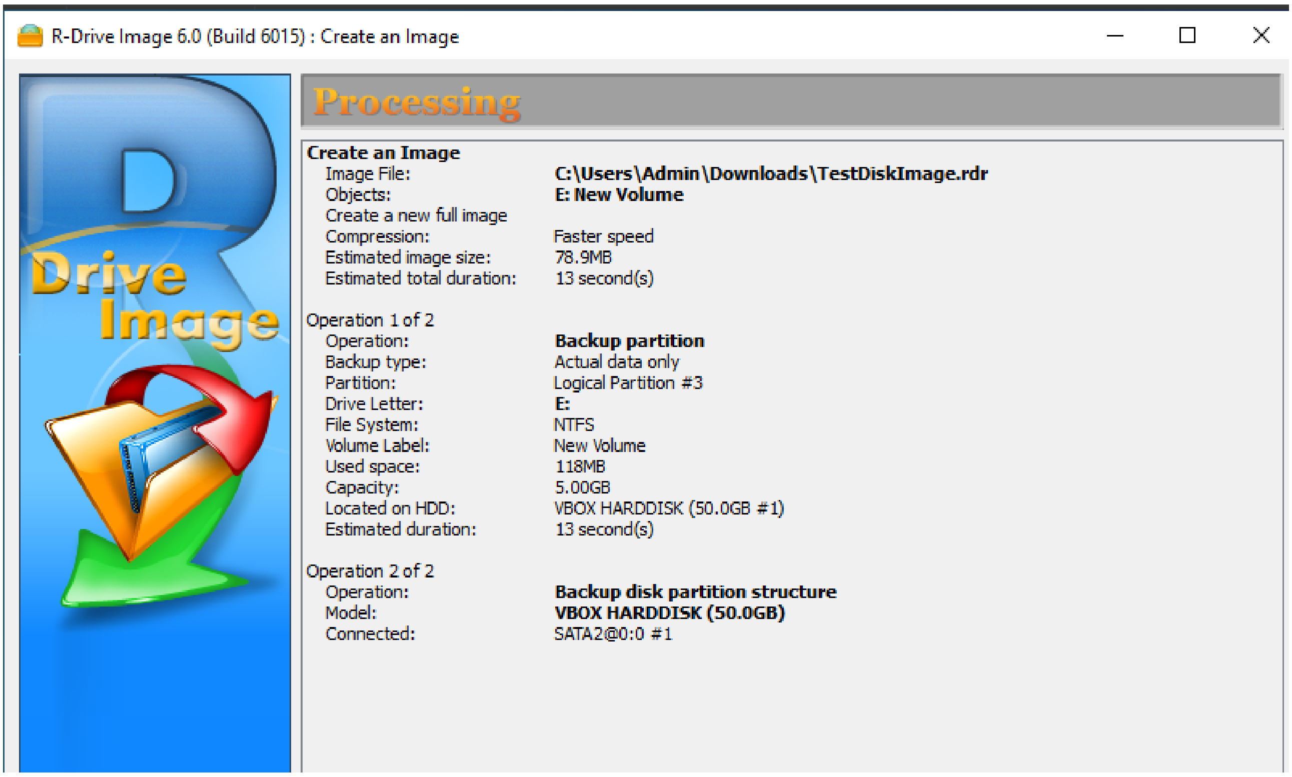


5. In image option click next



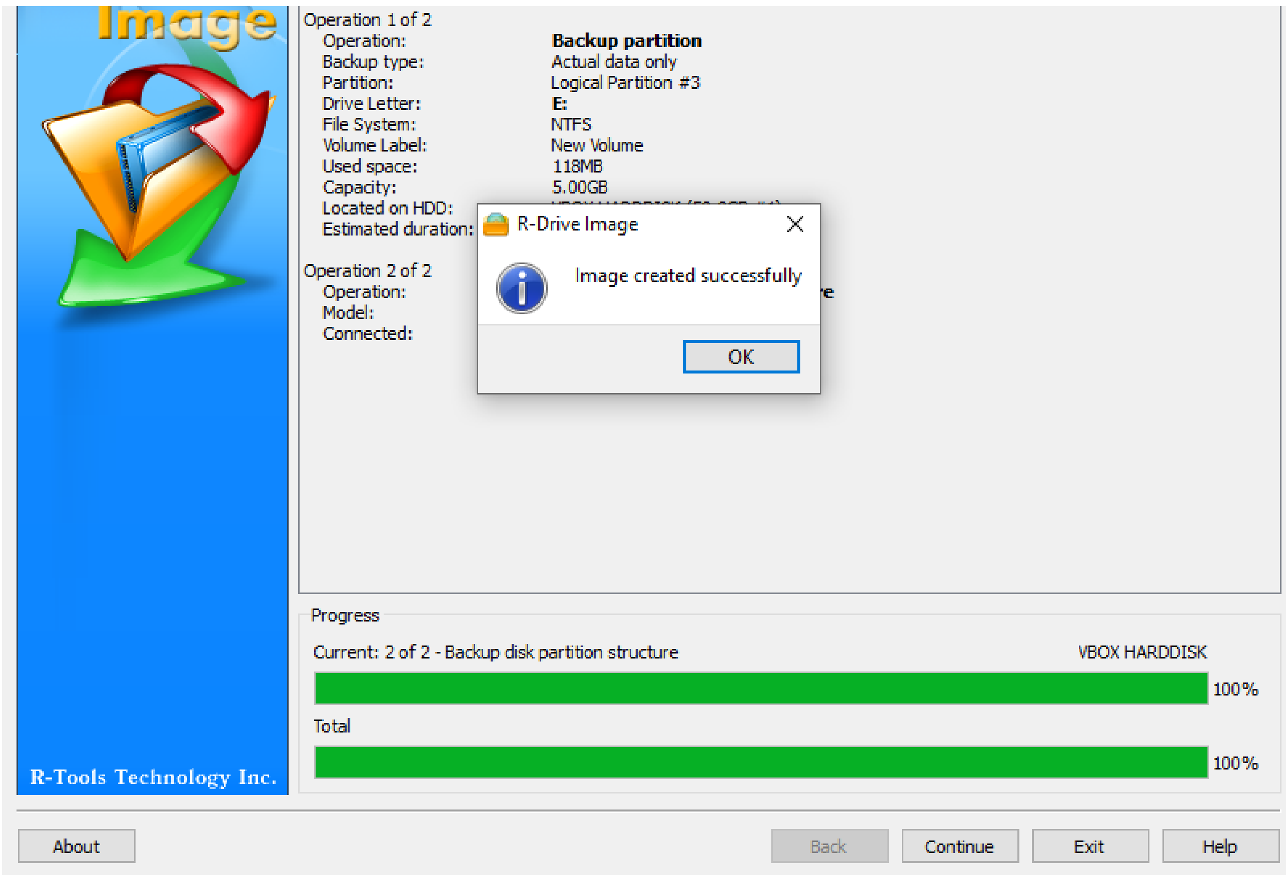
6. In the backup options window click next





7. The progress bar in the processing window will show the completed percentage task.

8. Once the process is done the following pop-up window is displayed , click ok.



9. Go to the saved location to view the created disk partition image file.